

Assignment 22

Q1. Create a login application using PreparedStatement to check username and password.

Java Code:

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.util.Scanner;

public class Q1_LoginApp {

    private static final String URL = "jdbc:mysql://localhost:3306/SIT_CSE";
    private static final String USER = "root";
    private static final String PASSWORD = "root";

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Username: ");
        String username = sc.nextLine();

        System.out.print("Enter Password: ");
        String password = sc.nextLine();

        String query = "SELECT * FROM users WHERE username = ? AND password = ?";

        try (Connection con = DriverManager.getConnection(URL, USER, PASSWORD);
            PreparedStatement ps = con.prepareStatement(query)) {

            // Setting parameters using PreparedStatement
            ps.setString(1, username);
            ps.setString(2, password);
```

```

        // Executing query and processing ResultSet
        try (ResultSet rs = ps.executeQuery()) {
            if (rs.next()) {
                System.out.println("\nLogin Successful! Welcome, " +
rs.getString("username") + ".");
            } else {
                System.out.println("\nLogin Failed! Invalid Username or
Password.");
            }
        }

    } catch (Exception e) {
        System.out.println("Database Error: " + e.getMessage());
    } finally {
        sc.close();
    }
}
}

```

SQL Code:

```
USE SIT_CSE;
```

```

CREATE TABLE IF NOT EXISTS users (
    id INT PRIMARY KEY AUTO_INCREMENT,
    username VARCHAR(50) UNIQUE NOT NULL,
    password VARCHAR(50) NOT NULL
);

```

```
-- Sample Data
```

```

INSERT INTO users (username, password) VALUES
    ('admin', 'admin123'),
    ('akshat', 'pass123')
ON DUPLICATE KEY UPDATE username = VALUES(username);

```

Output:

```
Enter Username: akshat
Enter Password: pass123

Login Successful! Welcome, akshat.
```

```
Enter Username: akshat
Enter Password: wrongpass

Login Failed! Invalid Username or Password.
```

Q2. Create a Java application using PreparedStatement to authenticate hospital staff login (doctor/nurse/login ID and password). Display appropriate access messages based on authentication result

Java Code:

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.util.Scanner;

public class Q2_HospitalLoginApp {

    private static final String URL = "jdbc:mysql://localhost:3306/SIT_CSE";
    private static final String USER = "root";
    private static final String PASSWORD = "root";

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.println("===== HOSPITAL STAFF LOGIN
=====");
        System.out.print("Enter Login ID: ");
        String loginId = sc.nextLine();

        System.out.print("Enter Password: ");
```

```

String password = sc.nextLine();

String query = "SELECT name, role FROM hospital_staff WHERE
login_id = ? AND password = ?";

try (Connection con = DriverManager.getConnection(URL, USER,
PASSWORD);
    PreparedStatement ps = con.prepareStatement(query)) {

    ps.setString(1, loginId);
    ps.setString(2, password);

    try (ResultSet rs = ps.executeQuery()) {
        if (rs.next()) {
            String name = rs.getString("name");
            String role = rs.getString("role");

            System.out.println("\nAUTHENTICATION SUCCESSFUL!");
            System.out.println("Welcome, " + name + " (" + role + ")");

            // Role-based access message
            if ("Doctor".equalsIgnoreCase(role)) {
                System.out.println("Access Granted: Medical Records &
Prescriptions Portal.");
            } else if ("Nurse".equalsIgnoreCase(role)) {
                System.out.println("Access Granted: Patient Care & Ward
Monitoring Portal.");
            } else {
                System.out.println("Access Granted: General Staff Portal.");
            }
        } else {
            System.out.println("\nAUTHENTICATION FAILED!");
            System.out.println("Access Denied: Invalid Login ID or
Password.");
        }
    }

} catch (Exception e) {
    System.out.println("Database Error: " + e.getMessage());
} finally {

```

```
        sc.close();
    }
}
```

SQL Code:

```
USE SIT_CSE;
```

```
CREATE TABLE IF NOT EXISTS hospital_staff (
    login_id VARCHAR(50) PRIMARY KEY,
    password VARCHAR(50) NOT NULL,
    name VARCHAR(100) NOT NULL,
    role VARCHAR(20) NOT NULL -- Role: 'Doctor' or 'Nurse'
);
```

-- Sample Data

```
INSERT INTO hospital_staff (login_id, password, name, role) VALUES
    ('DOC101', 'docpass123', 'Dr. Sharma', 'Doctor'),
    ('NUR201', 'nursepass', 'Sister Mary', 'Nurse')
ON DUPLICATE KEY UPDATE name = VALUES(name);
```

Output:

Case 1: Doctor Login

```
===== HOSPITAL STAFF LOGIN =====
Enter Login ID: DOC101
Enter Password: docpass123

AUTHENTICATION SUCCESSFUL!
Welcome, Dr. Sharma (Doctor)
Access Granted: Medical Records & Prescriptions Portal.
```

Case 2: Nurse Login

```
===== HOSPITAL STAFF LOGIN =====  
Enter Login ID: NUR201  
Enter Password: nursepass  
  
AUTHENTICATION SUCCESSFUL!  
Welcome, Sister Mary (Nurse)  
Access Granted: Patient Care & Ward Monitoring Portal.
```

Case 3: Invalid Login

```
===== HOSPITAL STAFF LOGIN =====  
Enter Login ID: DOC101  
Enter Password: wrongpassword  
  
AUTHENTICATION FAILED!  
Access Denied: Invalid Login ID or Password.
```